Marine Technical Services Ltd

MTS Marine Technical Services Ltd

Lubieszyńska 5
72-006 Mierzyn, Poland
Phn: +48 91 852 48 18
Fax: +48 91 886 59 42
E-mail: office@mts.szczecin.pl
www.mts.szczecin.pl
Regon 320551471
VAT EU: PL9552237992

MV MOROCCO STAR
ME no. 2 SB OVERHAUL
31.03 – 16.06.2026

REPORT

SAMPLE

PREPARED BY
Marine Technical Services Ltd.
Szczecin, Poland

CERTIFIED
ISO 9001

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MTS Marine Technical Services Ltd
Lubieszyńska 5
72-006 Mierzyn, Poland
Phn: +48 91 852 48 18
Fax: +48 91 886 59 42
E-mail: office@mts.szczecin.pl
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REMARKS

Vessel:	**MOROCCO STAR**	**Work no:**	**MTS/2026/083**
Date/place:	**31.03 – 16.06.2026**	**Machinery:**	**ME 2 SB**
Manufacturer:	**MAK**	**Type:**	**8M32C**
Serial Number:	**38236**	**Runing hours:**	**89090**
OVERHAUL			

ME no. 2 SB side MAK8M32C

Apply to cylinder heads:

Eight cylinder heads were removed from the engine and transported from the vessel to the shipyard workshop for overhaul and inspection.
All cylinder heads were completely disassembled and thoroughly cleaned.

The following work was carried out on all cylinder heads:

- O-rings on the injector sleeves were renewed.
- Inlet and exhaust valve seats were machined and lapped to restore proper sealing conditions.
- On all cylinder heads the starting valves were internally cleaned, inspected and fitted with new gaskets.

Due to cylinder head no. 4 being found with internal crack it was scrapedand replaced with another sourced from vessel spare units.

The cylinder heads were subjected to pressure testing at 10 bar.
Testing of all cylinder heads was completed with satisfactory results, confirming adequate sealing integrity and suitability for continued service.

Valve clearances were adjusted as required.

Rocker arm bushes were renewed, restoring the correct bearing clearances.

Wear ridges on rocker arm pins were removed using a stone to eliminate surface irregularities and restore smooth operating surfaces. Final polishing of the recovered pins was completed to achieve the required finish and improve long-term reliability
The rocker arms were assembled on the engine, aligned and properly secured.
The cylinder heads were fully armed and installed on the engine.

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Final tightening of the cylinder head studs was carried out in accordance with the manufacturer's specifications.

Apply to cylinder liners:

All cylinder liners, 8 pcs. were pulled out from the Engine and transported from the vessel to the shipyard workshop for overhaul and inspection.
All cylinder liners were cleaned, inspected and honed.
Cylinder liners nos. 2, 4, 5, 6, 7and 8 were calibrated.
Cylinder liners nos. 1 and 3 were replaced with new ones .
After completion of the workshop overhaul, the cylinder liners were reinstalled on new seals including correct positioning and verification of proper seating.

Apply engine block:

The engine block was cleaned.
Hand lapping of the cylinder liner landing surfaces on the engine block was carried out to improve the surface finish and ensure correct seating conditions for the cylinder liners during installation.

Apply to cooling jackets:

Cooling jackets were dismounted from the engine block and transported from the vessel to the shipyard workshop for overhaul and inspection.
Cooling jackets were cleaned and freed from scale, corrosion products and deposits.
The cooling jackets were reassembled to the engine.

Apply to pistons:

8 pcs of pistons were dismounted from the engine and transported from the vessel to the shipyard workshop for overhaul and inspection .
All pistons were measured.

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72-006 Mierzyn, Polska
Phn: +48 91 852 48 18
Fax: +48 91 886 59 42
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Piston No. 3 was completely disassembled and its interior—including the cooling spaces—was inspected and cleaned, after which it was reassembled the piston interior condition found in very good condition.
Gudgeon pins were measured and found in good condition.
All pistons
were reassembled to the engine.

Apply to connecting rods:

8 pcs. of connecting rods were disassembled from the engine and and transported from the vessel to the shipyard workshop for overhaul and inspection.
Connecting rods were cleaned and verified. The connecting rod no. 1 big end bore was machined by an external service measurement. Results attached.
Connecting rods were reassembled on the engine with new oversized big end bearings.
All bolts were checked and tightened according to producer manual.

Apply to crankshaft:
Crankpins were cleaned, verified and measured .
Due to diameter and ovality were exceeded maker´s tolerances crankpins no.1, no.2, no.3, no.4, no.5 and no.7 were machined- by the external service company.
Blue-check (blueprinting) of crankshaft crankpins were carried out .
Final polishing was carried out to achieve the required surface finish.
Routhness of the pins surfaces was measured and found wihtin the acceptable limits.
The crack inspection was carried out using the Magna Flux method.
No cracks or surface defects were detected.

Crankcase was cleaned and verified.

Main and thrust bearings were replaced for new ones.

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OVERHAUL			

Apply to camshaft:

All camshaft sections were visually inspected and found in a fair to good visual condition.
Clearances between camshaft pins and camshaft bearings were measured and found within Producer limits.

Apply to fuel oil pump

Fuel oil pumps were dismantled from the engine , transferred from the vessel to the shipyard workshop.
Fuel oil pumps were cleaned and inspected .
Fuel oil pump elements were replaced for new ones .
Overhaul of two fuel oil pumps (FOPs) was carried out using pump bodies sourced from vessel spare units (used condition). Work included inspection, cleaning and reassembly to restore operational performance.
All pumps assembled on new seals.
Fuel oil pump were reassembled to the engine.

Apply to fuel injection valves:

Fuel injection valves were disassembled, cleaned and verified.
Fuel injection valves were assembled on the engine with new nozzles and sealings.
Adjustment and calibration of 8 pcs. fuel injectors was carried out with opening pressure set and verified at 450 bar.

Apply to cooling water system:

On the fresh water (FW) cooling system new gaskets and O-rings were instaled.
The fresh water (FW) cooling system was filled and pressure-tested at 2.5 bar.
Within the achieved test pressure the system was confirmed tight with no leakage detected.

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Apply to Engine Running tests:
Start-up tests carried out with engine running without load for 1, 5, and 15 minutes.
After each interval, the crankcase was opened and temperatures of main and connecting rod bearings were measured.

Other works carried out :

- Crankshat deflection x3 tiimes, before overhaul, after overhaul (cold) – included in attached, after overhaul in hot condition after running tests
- Replacement of rocker arm bearing bushes was carried out on additional valve rocker assemblies intended as spare units for vessel stock ensuring serviceable backup components are available onboard for future maintenance requirements.
- Replacement of O-rings on crankcase covers, timing gear covers and valve covers was carried out
- Tested low LO pressure shutdown and emergency stop from the local control station

All screws were tightened according to producer manual.
Engine cleaned, started, engine parameters controlled.
Engine working was passed to the Vessel Crew.

Recommendations:

- Valve clearances adjusting checking and readjusting according to the producer manual.
- Cylinder heads bolts tightening checking and retightening according to producer manual.
- Connecting rods bolts tightening checking and retightening according to producer manual.
- Main bearing covers bolts tightening checking and retightening according to producer manual.
- All other screws tightening checking and retightening according to producer manual.

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OVERHAUL			

Additional works and sea trials :

Just before proceeding with the sea trials, insufficient lubrication of the ME 2 SB and ME 2 PS rocker arms was discovered.

It was found that the bushes in the rocker arms had initially been installed with an incorrect orientation (rotated by 180°), which caused insufficient oil supply to the valve bridges and, consequently, inadequate lubrication of the valves.

All rocker arm bushes were reinstalled in their correct positions, which resolved this issue.

During subsequent checks with the engines running, it was observed that on three units of ME 2 PS and three units of ME 2 SB, one inlet valve on each unit was still receiving insufficient oil supply.

As per the shipowner's decision, all rocker arms installed on the ME 2 SB engine were replaced with rocker arms transferred from the ME 1 SB engine. Similarly, the rocker arms on the ME 2 PS engine were replaced with rocker arms transferred from the ME 1 PS engine.

Following the completion of the repair work on ME 2 SB and ME 2 PS, the vessel proceeded to sea trials.

All required operational measurements were carried out during the sea trials, and the recorded results are attached to this report.

The additional work required to restore proper lubrication of the rocker arms on both engines delayed the commencement of the sea trials by six hours.

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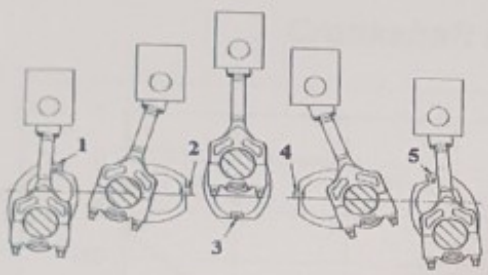
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MEASUREMENTS OF CRANKSHAFT DEFLECTION REPORT

Vessel:	MOROCCO STAR	Work no:	MTS/2026/083
Date/place:	31.03- 16.06.2026	Machinery:	ME2 SB
Manufacturer:	MAK	Type:	8 M32
Serial No.:	38236	Working hours:	89090
Reason of attendance:	OVERHAUL		



Crankcase Temp : [ ] °C

Draft Fwd : [ ] Draft Aft : [ ]

Cyl. no.	Crank deflection in 1/100mm (0,01mm)							
	1	2	3	4	5	6	7	8
(1) bottom start -X	0	0	0	0	0	0	0	0
(2) Port Side - P	-3	-1	1	1	1	1	-1	0
(3) TOP -T	-7	-2	3	4	2	2	1	1
(4) Starboard side S	-4	0	2	3	1	2	1	1
(5) Bottom stop - Y	0	0	0	2	0	1	0	0
Bottom Total - B=0.5*(X+Y)	0	0	0	1	0	0,5	0	0
Vertical deflection V=T-B	-7	-2	3	3	2	1,5	1	1
Horizontal deflection H=P-S	-1	-1	-1	-2	0	-1	-2	-1
Gauge reading C=T+B	-7	-2	3	5	2	2,5	1	1
Gauge reading D=P+S	-7	-1	3	4	2	3	0	1

If readings are OK Values C and D should be almost same

Measurements:					Limits:
Engine:	Before overhaul	[x]	After repairs	[ ]	
Measurements taken:	Cold	[x]	Warm	[ ]	
	Flywheel side	[ ]	Non-flywheel side	[ ]	all cylinders - (+5) - (-8)
Measured by:	Kraczkowski D.				
Remarks:	Crankshaft deflection readings taken using a certified vessel dial gauge.				

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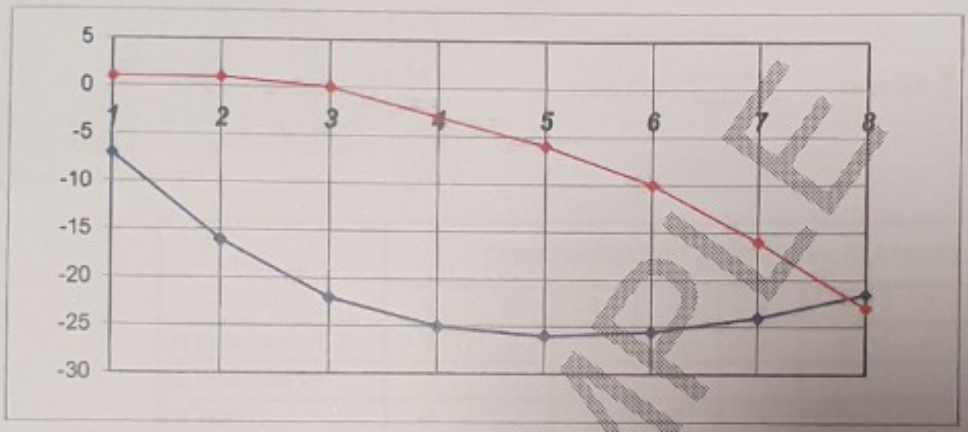
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72-006 Mierzyn, Poland
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E-mail: office@mts.szczecin.pl
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Vessel:	**MOROCCO STAR**	**Work no:**	**MTS/2026/083**
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Manufacturer:	**MAK**	**Type:**	**8 M32**
Serial No.:	**38236**	**Working hours:**	**89090**
Reason of attendance:	**OVERHAUL**		

Crankshaft Deflections -GRAPH



Vertical deflection (top -bottom)

Horizontal Deflection (Port -Starboard)

SAMPLE

Measurements:	Before overhaul	[x]	After repairs	[ ]	**Limits:**
Engine:	Cold	[x]	Warm	[ ]	
Measurements taken:	Flywheel side	[ ]	Non-flywheel side	[ ]	**all cylinders - (+5) - (-8)**
Measured by:	Kraczkowski D.				
Remarks:	Crankshaft deflection readings taken using a certified vessel dial gauge.				

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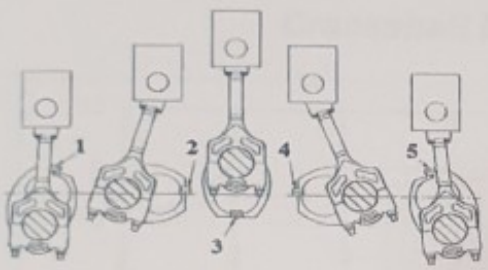
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Reason of attendance:	OVERHAUL		



Crankcase Temp : [ ] °C

Draft Fwd : [ ] Draft Aft : [ ]

Cyl. no.	Crank deflection in 1/100mm (0,01mm)							
	1	2	3	4	5	6	7	8
(1) bottom start -X	0	0	0	0	0	0	0	0
(2) Port Side - P	-2	-1	1	1	-1	1	-1	-2
(3) TOP -T	-4	-2	2	2	1	2	1	-3
(4) Starboard side S	-3	-1	1	1	-1	1	1	-2
(5) Bottom stop - Y	0	0	0	0	0	0	0	0
Bottom Total - B=0.5*(X+Y)	0	0	0	0	0	0	0	0
Vertical deflection V=T-B	-4	-2	2	2	1	2	1	-3
Horizontal deflection H=P-S	1	0	0	0	0	0	-2	0
Gauge reading C=T+B	-4	-2	2	2	1	2	1	-3
Gauge reading D=P+S	-5	-2	2	2	-2	2	0	-4

If readings are OK Values C and D should be almost same

Measurements:	Before overhaul	[ ]	After repairs	[x]	Limits:
Engine:	Cold	[x]	Warm	[ ]	
Measurements taken:	Flywheel side	[ ]	Non-flywheel side	[ ]	all cylinders - (+5) - (-8)
Measured by:	Kraczkowski D.				
Remarks:	Crankshaft deflection readings taken using a certified vessel dial gauge.				

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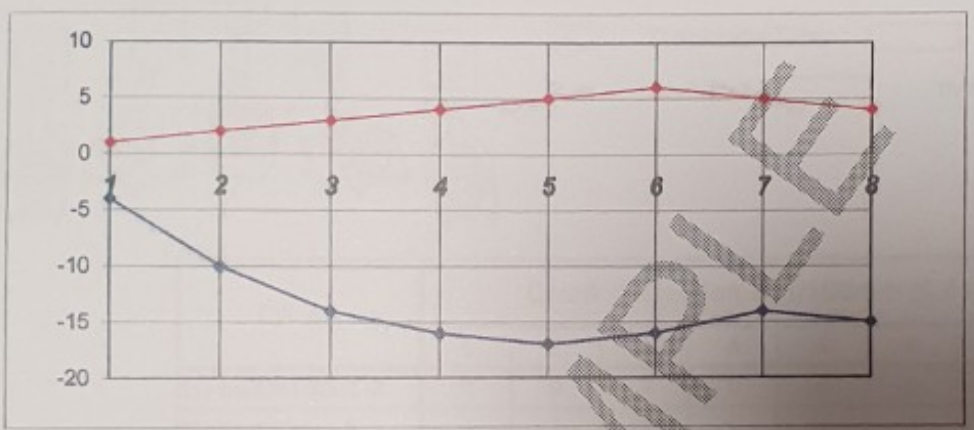
Lubieszyńska 5
72-006 Mierzyn, Poland
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Serial No.:	38236	Working hours:	89090
Reason of attendance:	OVERHAUL		

Crankshaft Deflections -GRAPH



Vertical deflection (top -bottom)

Horizontal Deflection (Port -Starboard)

SAMPLE

Measurements:	Before overhaul	☐	After repairs	☒	Limits:
Engine:	Cold	☒	Warm	☐	
Measurements taken:	Flywheel side	☐	Non-flywheel side	☐	all cylinders - (+5) - (-8)
Measured by:	Kraczkowski D.				
Remarks:	Crankshaft deflection readings taken using a certified vessel dial gauge.				

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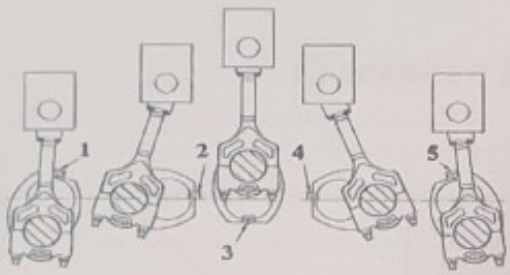
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Crankcase Temp : [ ] °C

Draft Fwd : [ ] Draft Aft : [ ]

Cyl. no.	Crank deflection in 1/100mm (0,01mm)							
	1	2	3	4	5	6	7	8
(1) bottom start -X	0	0	0	0	0	0	0	0
(2) Port Side - P	-1	1	1	1	1	1	1	1
(3) TOP -T	-3	-2	2	1	1	2	1	-2
(4) Starboard side S	1	-1	1	1	1	1	1	-1
(5) Bottom stop - Y	0	0	0	0	0	0	0	0
Bottom Total - B=0.5*(X+Y)	0	0	0	0	0	0	0	0
Vertical deflection V=T-B	-3	-2,0	2	1	1,0	2	1	-2
Horizontal deflection H=P-S	-2	2	0	0	0	0	0	2
Gauge reading C=T+B	-3	-2	2	1	1	2	1	-2
Gauge reading D=P+S	0	0	2	2	2	2	2	0

If readings are OK Values C and D should be almost same

Measurements:					Limits:
Engine:	Before overhaul	[ ]	After repairs	[x]	Cranks 1and 6 - (-10) - (+0)
Measurements taken:	Cold	[ ]	Warm	[x]	Cranks 2,3,4,5 - (-5) - (+5)
	Flywheel side	[ ]	Non-flywheel side	[ ]	
Measured by:	Stupakov M.				
Remarks:	Crankshaft deflection readings taken using a certified vessel dial gauge.				

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Crankshaft Deflections -GRAPH



Vertical deflection (top -bottom)

Horizontal Deflection (Port -Starboard)

SAMPLE

Measurements:	Before overhaul	☐	After repairs	☒	Limits:
Engine:	Cold	☐	Warm	☒	Cranks 1and 8 - (-10) - (+0)
Measurements taken:	Flywheel side	☐	Non-flywheel side	☐	Cranks 2,3,4,5 - (-5) - (+5)
Measured by:	Stupakov M.				
Remarks:	Crankshaft deflection readings taken using a certified vessel dial gauge.				

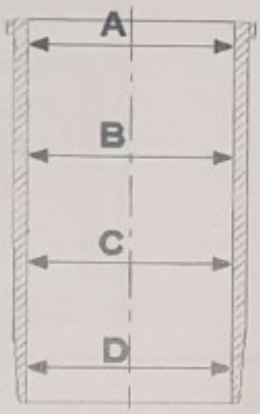
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MEASUREMENT REPORT OF THE LINERS

Vessel:	**MOROCCO STAR**	**Work no:**	**MTS/2026/083**
Date/place:	**31.03- 16.06.2026**	**Machinery:**	**ME2 SB**
Manufacturer:	**MAK**	**Type:**	**8 M32**
Serial No.:	**38236**	**Working hours:**	**89090**
Reason of attendance:	**OVERHAUL**		



Nominal:	**320**
Wear limit:	**1.00 mm**
Max. Oval:	**0.30 mm**
Measurements:	
Before honing	☐
After honing	☒

SAMPLE

Cylinder liners diameter [mm]

Cyl. No.		1	2	3	4	5	6	7	8
A	for-aft	X	320,14	X	320,15	320,10	320,15	320,30	320,15
A	stb-pb	X	320,04	X	320,15	320,10	320,15	320,30	320,15
B	for-aft	X	320,07	X	320,10	320,05	320,02	320,15	320,07
B	stb-pb	X	320,07	X	320,10	320,05	320,02	320,15	320,07
C	for-aft	X	320,05	X	320,05	320,00	320,00	320,10	320,01
C	stb-pb	X	320,05	X	320,05	320,00	320,00	320,10	320,01
D	for-aft	X	320,02	X	320,02	320,00	320,00	320,01	320,01
D	stb-pb	X	320,02	X	320,02	320,00	320,00	320,02	320,01
Max. Ovality			**0,10**		**0,00**	**0,00**	**0,00**	**0,01**	**0,00**

Measured:	Before overhaul ☐	After overhaul ☒	
Measured by:	Kraczkowski D.	**Measurement tool:**	inside micrometer MTS/A/072
Remarks:			

ISO 9001
BUREAU VERITAS
Certification

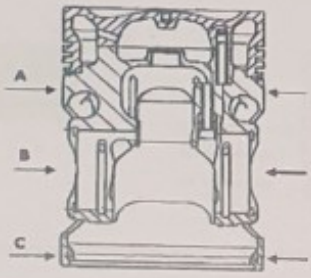
Wpis do Rejestru Przedsiębiorców prowadzonego przez Sąd Rejonowy Szczecin - Centrum w Szczecinie,
XIII Wydział Gospodarczy Krajowego Rejestru Sądowego pod numerem KRS 0000312044.
Wysokość kapitału zakładowego 51 000 PLN.

Marine Technical Services Ltd

MTS Marine Technical Services Ltd
Lubieszyńska 5
72-006 Mierzyn, Poland
Phn: +48 91 852 48 18
Fax: +48 91 886 59 42
E-mail: office@mts.szczecin.pl
www.mts.szczecin.pl
Regon 320551471
VAT EU: PL9552237992

MEASUREMENT REPORT OF THE PISTONS

Vessel:	MOROCCO STAR	Work no:	MTS/2026/083
Date/place:	31.03- 16.06.2026	Machinery:	ME2 SB
Manufacturer:	MAK	Type:	8 M32
Serial No.:	38236	Working hours:	89090
Reason of attendance:	OVERHAUL		





Nominal:	320

SAMPLE

Pistons diameter [mm]									
Cyl. No.		1	2	3	4	5	6	7	8
A	for-aft	319,70	319,70	319,70	319,70	318,71	319,70	319,70	319,70
A	stb-pb	318,90	318,90	318,90	318,91	318,91	318,90	318,91	318,90
B	for-aft	318,91	318,91	318,92	318,91	318,92	318,91	318,91	318,91
B	stb-pb	X	X	X	X	X	X	X	X
C	for-aft	319,70	319,72	318,71	319,70	318,71	319,70	319,70	319,70
C	stb-pb	318,91	318,91	318,92	318,92	318,93	318,91	318,92	318,92

Measured: Before overhaul ☐ After overhaul ☒

Measured by:	Kraczkowski D.	Measurement tool:	micrometer MTS/LP-300

Remarks:

Wpis do Rejestru Przedsiębiorców prowadzonego przez Sąd Rejonowy Szczecin - Centrum w Szczecinie, XIII Wydział Gospodarczy Krajowego Rejestru Sądowego pod numerem KRS 0000312044.
Wysokość kapitału zakładowego 51 000 PLN.

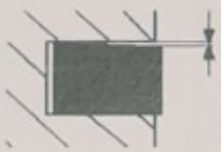
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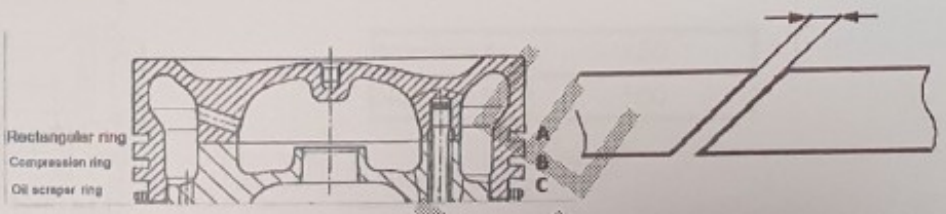
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72-006 Mierzyn, Poland
Phn: +48 91 852 48 18
Fax: +48 91 886 59 42
E-mail: office@mts.szczecin.pl
www.mts.szczecin.pl
Regon 320551471
VAT EU: PL9552237992

Marine Technical Services Ltd

MEASUREMENT REPORT OF THE PISTON RINGS

Vessel:	MOROCCO STAR	Work no:	MTS/2026/083
Date/place:	31.03- 16.06.2026	Machinery:	ME2 SB
Manufacturer:	MAK	Type:	8 M32
Serial No.:	38236	Working hours:	89090
Reason of attendance:	OVERHAUL		





Gap clearance
Unit no. in [mm]

No.	1	2	3	4	5	6	7	8
A	0,30	0,30	0,30	0,45	0,30	0,33	0,35	0,30
B	0,20	0,25	0,20	0,28	0,25	0,28	0,25	0,20
C	0,13	0,13	0,13	0,10	0,13	0,13	0,10	13,00

Measured:	Before overhaul ☐	After overhaul [x]	
Measured by:	Kraczkowski D.	Measurement tool:	feler gauge
Remarks:			

SAMPLE

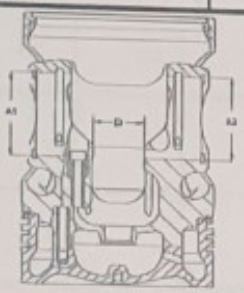
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Lubieszyńska 5
72-006 Mierzyn, Poland
Phn: +48 91 852 48 18
Fax: +48 91 886 59 42
E-mail: office@mts.szczecin.pl
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Regon 320551471
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MEASUREMENT REPORT OF THE PISTON HOLES

Vessel:	**MOROCCO STAR**	**Work no:**	**MTS/2026/083**
Date/place:	**31.03- 16.06.2026**	**Machinery:**	**ME2 SB**
Manufacturer:	**MAK**	**Type:**	**8 M32**
Serial No.:	**38236**	**Working hours:**	**89090**
Reason of attendance:	**OVERHAUL**		



Nominal A:	150
Nominal B:	150

Piston Holes A diameter [mm]									
Cyl. No.		1	2	3	4	5	6	7	8
A1	V	150,04	150,04	150,05	150,03	150,05	150,04	150,05	150,05
A1	H	150,04	150,04	150,05	150,04	150,05	150,05	150,05	150,05
A2	V	150,04	150,04	150,05	150,03	150,05	150,04	150,05	150,05
A2	H	150,04	150,04	150,05	150,04	150,04	150,05	150,05	150,05
Ovality		**0,00**	**0,00**	**0,00**	**0,01**	**0,01**	**0,01**	**0**	**0**
Bush diameter B [mm]									
B1	V	150,15	150,16	150,16	150,16	150,15	150,16	150,16	150,15
B1	H	150,16	150,18	150,18	150,18	150,17	150,18	150,18	150,17
Ovality		**0,01**	**0,02**	**0,02**	**0,02**	**0,02**	**0,02**	**0,02**	**0,02**
Boss distance D [mm]									
D		85,83	85,67	85,77	85,77	85,78	85,91	85,85	85,97
Bush width X [mm]									
X		84,95	84,97	84,93	84,92	84,97	84,95	84,97	84,95
CLEARANCES									
A-C [max 0.06]		0,01	0,01	0,02	0,01	0,02	0,02	0,02	0,05
B-C [max 0.25]		0,13	0,15	0,15	0,15	0,14	0,15	0,15	0,17
D-X [max 1.1]		0,88	0,70	0,84	0,85	0,81	0,96	0,88	1,02

Measured: Before overhaul ☐ After overhaul ☐

Measured by:	**Kraczkowski D.**	**Measurement tool:**	micrometer MTS/N/2021/2 and feeler gauge

Remarks:

SAMPLE

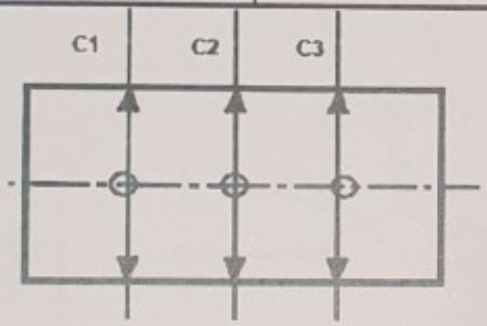
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MTS Marine Technical Services Ltd

Lubieszyńska 5
72-006 Mierzyn, Poland
Phn: +48 91 852 48 18
Fax: +48 91 886 59 42
E-mail: office@mts.szczecin.pl
www.mts.szczecin.pl
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VAT EU: PL9552237992

MEASUREMENT REPORT OF THE GUDGEON PINS

Vessel:	**MOROCCO STAR**	**Work no:**	**MTS/2026/083**
Date/place:	**31.03- 16.06.2026**	**Machinery:**	**ME2 SB**
Manufacturer:	**MAK**	**Type:**	**8 M32**
Serial No.:	**38236**	**Working hours:**	**89090**
Reason of attendance:	**OVERHAUL**		



Nominal C:	150

SAMPLE

Gudgeons pins diameter [mm]

Cyl. No.		1	2	3	4	5	6	7	8
C1	for-aft	150,03	150,03	150,03	150,03	150,03	150,03	150,03	150,00
C1	stb-pb	150,03	150,03	150,03	150,03	150,03	150,03	150,03	150,00
Ovality		**0,00**	**0,00**	**0,00**	**0,00**	**0,00**	**0,00**	**0,00**	**0,00**
C2	for-aft	150,03	150,03	150,03	150,03	150,03	150,03	150,03	150,00
C2	stb-pb	150,03	150,03	150,03	150,03	150,03	150,03	150,03	150,00
Ovality		**0,00**	**0,00**	**0,00**	**0,00**	**0,00**	**0,00**	**0,00**	**0,00**
C3	for-aft	150,03	150,03	150,03	150,03	150,03	150,03	150,03	150,00
C3	stb-pb	150,03	150,03	150,03	150,03	150,03	150,03	150,03	150,00
Ovality		**0,00**	**0,00**	**0,00**	**0,00**	**0,00**	**0,00**	**0,00**	**0,00**

Piston pin bush - piston pin **Minimum clearance: 0.111 mm** **Limit clearance:**
Maximum clearance: 0.160 mm **0.210 mm**

Measured: Before overhaul ☐ After overhaul ☒

Measured by:	Kraczkowski D.	**Measurement tool:**	micrometer MTS/A/072

Remarks:

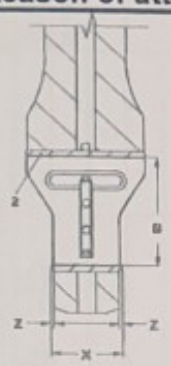
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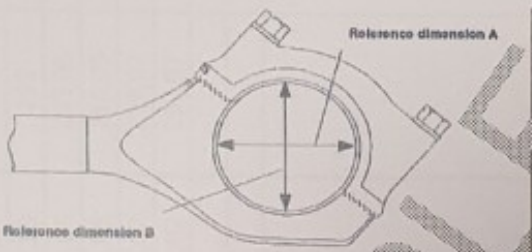
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Lubieszyńska 5
72-006 Mierzyn, Poland
Phn: +48 91 852 48 18
Fax: +48 91 886 59 42
E-mail: office@mts.szczecin.pl
www.mts.szczecin.pl
Regon 320551471
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MEASUREMENT REPORT OF THE CONNECTING RODS

Vessel:	MOROCCO STAR	Work no:	MTS/2026/083
Date/place:	31.03- 16.06.2026	Machinery:	ME2 SB
Manufacturer:	MAK	Type:	8 M32
Serial No.:	38236	Working hours:	89090
Reason of attendance:	OVERHAUL		





	big end bearing
Nominal:	(A) 280.25 +0.05
	(B) A + 0.04 + 0.02

	big end bearing
Nominal:	280
	Small end bearing
Nominal:	150
Oil clearance	0,16
Clearance Limit	0,21

Big end Inner Diameter [mm]							without bearing shells		
Cyl. No.		1	2	3	4	5	6	7	8
A	FWD	279,71	279,77	279,70	279,71	279,71	292,99	279,7	292,98
A	AFT	279,71	279,76	279,70	279,72	279,71	292,99	279,7	292,98
B	FWD	279,76	279,85	279,79	279,77	279,78	292,98	279,77	293,01
B	AFT	279,76	279,86	279,79	279,78	279,78	292,99	279,78	293,03
Ovality							0,01		

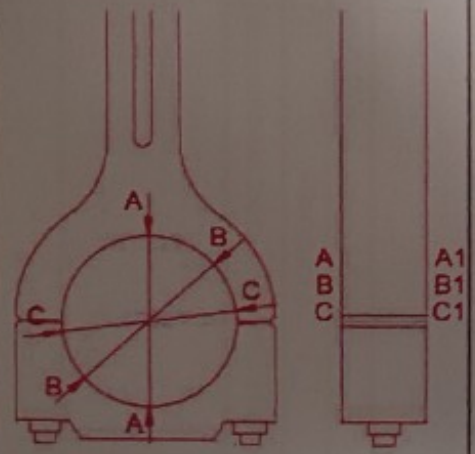
Small end diameter [mm]							with small end bush		
Cyl. No.		1	2	3	4	5	6	7	8
B1	FWD	150,15	150,16	150,16	150,16	150,15	150,16	150,16	150,15
B2	FWD	150,16	150,18	150,18	150,18	150,17	150,18	150,18	150,17
B1	AFT	150,15	150,16	150,16	150,16	150,15	150,16	150,16	150,15
B2	AFT	150,16	150,18	150,18	150,18	150,17	150,18	150,18	150,17
Ovality		0,01	0,02	0,02	0,02	0,02	0,02	0,02	0,02
Oil clearance(B-C)		0,13	0,15	0,15	0,15	0,14	0,15	0,15	0,17

Measured:	Before overhaul ☐	After overhaul ☐	
Measured by:	Kraczkowski D.	Measurement tool:	inside micrometer MTS/A/067, MTS/N/2021/2
Remarks:			

SAMPLE

CONNECTING ROD INSPECTION FOR MAK 32			PLANT SHIP: MOROCCO STAR			ENGINE TYPE: MAK 32		
			ENGINE Nº: 2 SB			RUNNING HOURS:		
			SIGN:			INSP. DATE: 29/04/2026		

CYLINDER Nº		2LB	2PB	3	4	5	6	7	8
CONNECTING ROD IDENT Nº									
RUNNING HOURS									
	A	293	292,99	293	292,99	293	293	292,99	293
	A1	293	292,99	292,99	292,99	293	293	293	292,99
	B	293,05	293,03	293,05	293,03	293,05	293,04	293,04	293,05
	B1	293,05	293,03	293,05	293,03	293,05	293,04	293,04	293,05
	C	293,09	293,06	293,08	293,07	293,07	293,08	293,06	293,06
	C1	293,09	293,06	293,08	293,07	293,07	293,08	293,07	293,06
OVALNESS DIFF. BETWEEN MIN/MAX									
CONDITION OF SERRATION									



SAMPLE

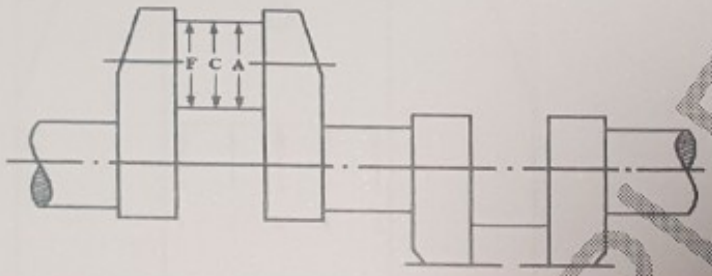
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Lubieszyńska 5
72-006 Mierzyn, Poland
Phn: +48 91 852 48 18
Fax: +48 91 886 59 42
E-mail: office@mts.szczecin.pl
www.mts.szczecin.pl
Regon 320551471
VAT EU: PL9552237992

Marine Technical Services Ltd

MEASUREMENT REPORT OF THE CRANK PINS

Vessel:	**MOROCCO STAR**	**Work no:**	**MTS/2026/083**
Date/place:	**31.03- 16.06.2026**	**Machinery:**	**ME2 SB**
Manufacturer:	**MAK**	**Type:**	**8 M32**
Serial No.:	**38236**	**Working hours:**	**89090**
Reason of attendance:	**OVERHAUL**		



Nominal: Ø	280

No.	Crank pins Unit no. in [mm]ø											
	1		2		3		4		5		6	
	Vertical	Horizontal	Vertical	Horizontal	Vertical	Horizontal	Vertical	Horizontal	Vertical	Horizontal	Vertical	Horizontal
Fore	279,96	279,90	279,91	279,87	279,96	279,91	279,97	279,90	279,95	279,90	277,96	277,96
Center	279,96	279,92	279,91	279,88	279,96	279,92	279,97	279,91	279,95	279,92	277,97	277,97
Aft	279,96	279,90	279,91	279,88	279,96	279,91	279,97	279,90	279,95	279,90	277,96	277,96
Oval	0,06		0,05		0,05		0,07		0,05		0,00	

No.	7		8	
	Vertical	Horizontal	Vertical	Horizontal
Fore	279,93	279,89	277,97	277,96
Center	279,93	279,89	277,97	277,96
Aft	279,93	279,88	277,97	277,96
Oval	0,05		0,01	

Axial clearance S=S1+S2 [0.30-0.50 mm - max 0.8 mm]

Measured:	Before overhaul [X]	After overhaul [ ]	
Measured by:	**Kraczkowski D.**	**Measurement tool:**	micrometer MTS/N/350
Remarks:			

COPY

Worldwide on site machining - crankshaft repair -
With mobile machine tools- spare parts - consulting

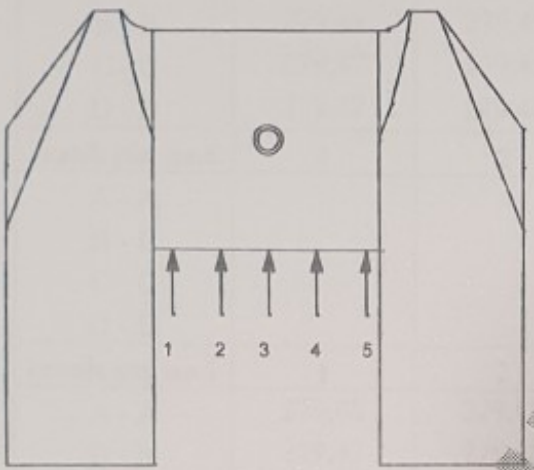
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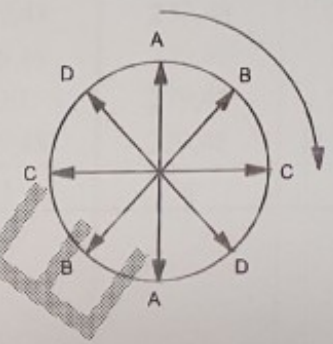
MTS
Kasulke & Rohde GmbH
Mobil Technik Service

MEASUREMENTS AFTER MACHINING

Schiffsname/ Vessel MV "Morocco Star"
Werk / Power plant : IMO: **7803190**

Datum/ Date: 17.05.202





Drehrichtung im Uhrzeigersinn vom Schwungrad gesehen
Clockwise direction from flywheel side

SAMPLE

Engine no.: MAK 8 M 32 C / Serial No. 38236 M.E. #2 STARBOARD

crank pin no.1	1	2	3	4	5
A – A	279,48	279,49	279,49	279,48	279,48
B - B	279,49	279,49	279,49	279,48	279,48
C - C	279,49	279,49	Oil	279,49	279,48
D - D	279,48	279,49	279,49	279,49	279,48
crank pin no.2	**1**	**2**	**3**	**4**	**5**
A - A	279,49	279,49	279,49	279,49	279,49
B - B	279,49	279,49	279,49	279,49	279,49
C - C	279,49	279,48	Oil	279,48	279,49
D - D	279,48	279,49	279,49	279,48	279,48
crank pin no.3	**1**	**2**	**3**	**4**	**5**
A - A	279,48	279,48	279,48	279,48	279,48
B - B	279,48	279,48	279,48	279,48	279,47
C - C	279,49	279,49	Oil	279,48	279,48
D - D	279,48	279,48	279,48	279,48	279,47
crank pin no.4	**1**	**2**	**3**	**4**	**5**
A - A	279,48	279,49	279,49	279,49	279,49
B - B	279,49	279,49	279,48	279,49	279,49
C - C	279,49	279,49	Oil	279,49	279,49
D - D	279,49	279,49	279,49	279,48	279,49

Worldwide on site machining - crankshaft repair -
With mobile machine tools- spare parts - consulting

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Phone: +49 40 79 68 61- 0 · Telefax: +49 40 79 68 61- 50
E-Mail: info@mts-insitu.com · Internet: www.mts-insitu.con

MTS
Kasulke & Rohde GmbH
Mobil Technic Service

After Machining

Engine No.: **MAK 8 M 32 C/Serial No.38236 M.E.#2 STBD** Date:. 17.05.2026

crank pin no.5	**1**	**2**	**3**	**4**	**5**
A - A	279,47	279,48	279,48	279,48	279,48
B - B	279,48	279,49	279,49	279,49	279,49
C - C	279,47	279,48	Oil	279,48	279,48
D - D	279,47	279,48	279,48	279,48	279,48
crank pin no.6	**1**	**2**	**3**	**4**	**5**
A - A					
B - B					
C - C					
D - D					
crank pin no.7	**1**	**2**	**3**	**4**	**5**
A - A	279,48	279,49	279,49	279,49	279,49
B - B	279,47	279,48	279,49	279,49	279,48
C - C	279,48	279,48	Oil	279,49	279,48
D - D	279,48	279,49	279,48	279,49	279,48
crank pin no.8	**1**	**2**	**3**	**4**	**5**
A - A					
B - B					
C - C					
D - D					
crank pin no.9	**1**	**2**	**3**	**4**	**5**
A - A					
B - B					
C - C					
D - D					

SAMPLE

Bemerkungen: Diameter dimensions after machining the crankpins number #1,#2,#3,#4,#5,#7 to
Remarks : minus 0,5mm.

Cernaval Shipyard, Algeciras, Spain

Reparaturort / Place of repair

Unterschrift / Sign. Serv. Eng.

Unterschrift Kunde / Signature customer

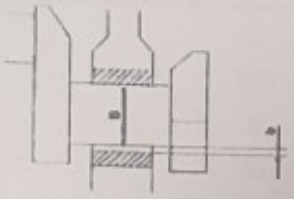
Marine Technical Services Ltd

MTS Marine Technical Services Ltd

Lubieszyńska 5
72-006 Mierzyn, Poland
Phn: +48 91 852 48 18
Fax: +48 91 886 59 42
E-mail: office@mts.szczecin.pl
www.mts.szczecin.pl
Regon 320551471
VAT EU: PL9552237992

MEASUREMENT REPORT OF CAMSHAFT BEARINGS

Vessel:	**MOROCCO STAR**	**Work no:**	**MTS/2026/083**
Date/place:	**31.03- 16.06.2026**	**Machinery:**	**ME2 SB**
Manufacturer:	**MAK**	**Type:**	**8 M32**
Serial No.:	**38236**	**Working hours:**	**89090**
Reason of attendance:	**OVERHAUL**		



Clearance between camshaft pins and camshaft bearings [mm]									
No.	1	2	3	4	5	6	7	8	9
Clearance	0,18	0,15	0,15	0,15	0,15	0,13	0,15	0,15	0,13

Measured:	Before overhaul ☐	After overhaul ☒	
Measured by:	Kraczkowski D.	**Measurement tool:**	feeler gauge
Remarks:			

SAMPLE

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TRIALS

Vessel:	**MOROCCO STAR**	**Work no:**	**MTS/2026/083**
Date/place:	**31.03- 16.06.2026**	**Machinery:**	**ME2 SB**
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Reason of attendance:	**OVERHAUL**		

Revolution	500	RPM
Oil pressure	6	bar
Oil temperature	52	°C
Fresh water pressure	3,5	bar
Fresh water temperature	42	°C
Pressure charging	0	bar
Fuel pressure	4,5	bar

SAMPLE

Main bearings temperatures [Celsius]										
Time	1	2	3	4	5	6	7	8	9	10
1 min	52	52	55	55	55	55	55	54	53	
5 min	56	57	57	56	56	56	56	55	55	
15 min	59	56	58	58	58	58	56	58	56	

Big-end bearings temperatures [Celsius]								
Time	1	2	3	4	5	6	7	8
1 min	53	54	52	55	54	54	55	55
5 min	57	56	56	56	56	56	56	56
15 min	58	57	57	57	57	56	58	57

Filled by :	Kraczkowski D.
Remarks:	

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TRIALS

Vessel:	**MOROCCO STAR**	**Work no:**	**MTS/2026/083**
Date/place:	**8.07- 9.07.2026**	**Machinery:**	**ME2 SB**
Manufacturer:	**MAK**	**Type:**	**8 M32**
Reason of attendance:	SEA TRIALS		

Load	2328	KW	78%
Cooling water temperature:	77	°C	
Oil pressure:	5,3	BAR	
Oil temperature	61	°C	
Engine speed:	600	RPM	
Fresh water pressure:	6,3	BAR	
Fresh water temperature:	76,0	°C	
Fuel pressure:	5,0	BAR	

SAMPLE

STARBOARD SIDE

Main bearings temperatures [Celsius]								
	1	2	3	4	5	6	7	8
-	63	65	64	65	62	63	64	64
Exhaust temperatures [Celsius]								
	1	2	3	4	5	6	7	8
	463	462	442	439	434	416	449	442
Minimum Exhaust gas pressure [Bar]								
	1	2	3	4	5	6	7	8
	122	119	116	119	119	120	117	117
Maximum Exhaust gas pressure [Bar]								
	1	2	3	4	5	6	7	8
	126	122	119	123	122	124	122	123

Filled by :	Stuoakov M.
Remarks:	

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NEW JOURNAL BEARING SHELLS ID MARKS

Vessel:	**MOROCCO STAR**	**Work no:**	**MTS/2026/083**
Date/place:	**31.03- 16.06.2026**	**Machinery:**	**ME2 SB**
Manufacturer:	**MAK**	**Type:**	**8 M32**
Serial No.:	**38236**	**Working hours:**	**89090**
Reason of attendance:	**OVERHAUL**		

New crankshaft journal Bearing Shells ID marks

UNIT No		Main Bearing Shells ID Marks
1	Top	3025CAT9124047041
	Bottom	3225CAT9124047075
2	Top	3025CAT9124047050
	Bottom	3225CAT9124047023
3	Top	3025CAT9124047047
	Bottom	3225CAT9124047012
4	Top	3025CAT9124047062
	Bottom	3225CAT9124047025
5	Top	3025CAT9124047004
	Bottom	3225CAT9124047016
6	Top	3025CAT9124047058
	Bottom	3225CAT9124047085
7	Top	3025CAT9124047006
	Bottom	3225CAT9124047050
8	Top	3025CAT9124047035
	Bottom	3225CAT9124047018
9	Top	3025CAT9124047051
	Bottom	3225CAT9124047005

Measured by:	Kraczkowski D.
Filled by:	
Remarks:	

SAMPLE

MV MOROCCO STAR MTS/2026/083
CYLINDER HEADS











MV MOROCCO STAR MTS/2026/083
CYLINDER LINERS,
COOLING JACKET CONDITION AND INSTALING









SAMPLE

MV MOROCCO STAR MTS/2026/083
PISTONS











MV MOROCCO STAR MTS/2026/083
FUEL INJECTION PUMP, FUEL INJECTION VALVE CONDITION
AND CONNECTING ROD MEASUREMENT









SAMPLE

MV MOROCCO STAR MTS/2026/083
CRANKPINS CONDITION
AND BLUE- PRINTING











MV MOROCCO STAR MTS/2026/083
ENGINE BLOCK



engine block condition



engine block condition



engine block liner landing surfaces machining



engine block liner landing surfaces lapping



engine block liner landing surfaces after lapping



engine parts instaling

SAMPLE